

Since publication of their article, the authors report no further potential conflict of interest.

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AA Amyloidosis and IgG4-Related Disease

TO THE EDITOR: Here, we describe a patient with renal amyloid A (AA) amyloidosis that was apparently associated with IgG4-related disease. A 53-year-old man with long-standing malaise and fatigue had a slow-growing mesenteric mass (5 cm in diameter) that had been present 16 years earlier on the basis of a review of previous imaging

studies. (The case had been included in a case series of IgG4-related disease.¹) In July 2015, the patient had received a diagnosis of mesenteric IgG4-related disease with involvement of surrounding lymph nodes, which had fulfilled international consensus criteria (Fig. 1A and 1B). The only blood abnormalities at that time had been

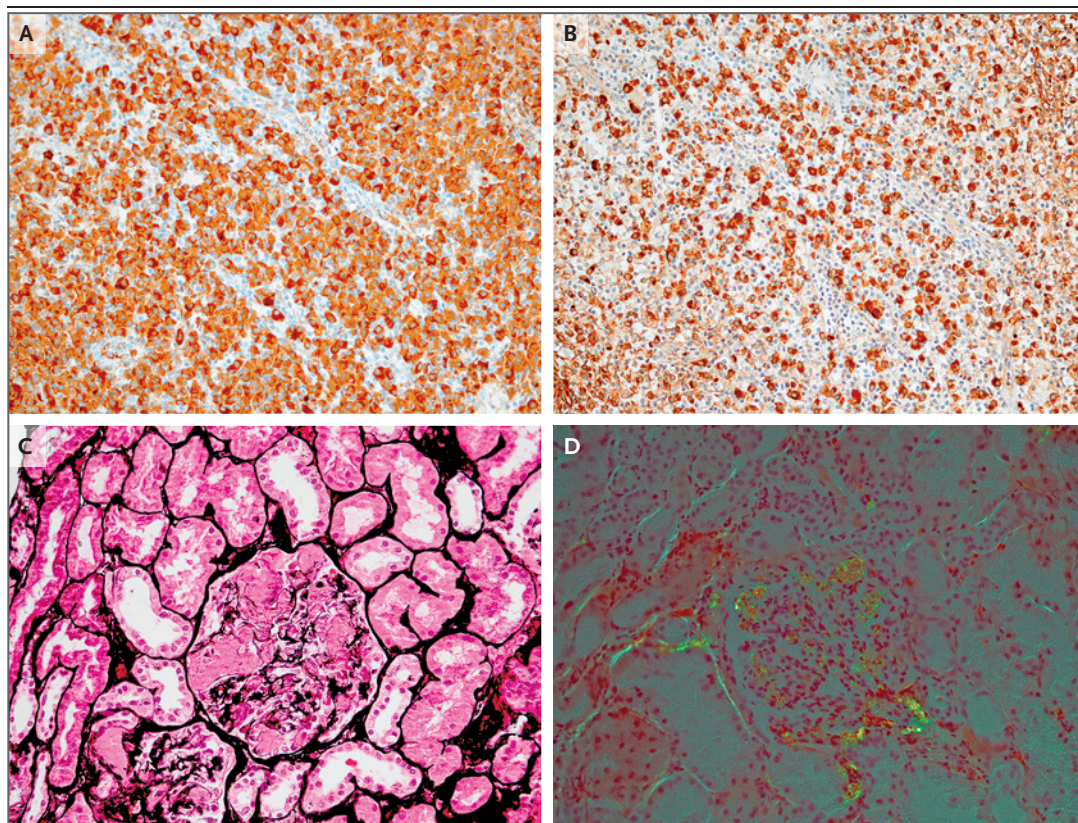


Figure 1. Mesenteric- and Renal-Biopsy Samples from the Patient.

Shown are biopsy samples obtained from a 53-year-old man with renal AA amyloidosis that was apparently associated with IgG4-related disease. A sample that was obtained from a mesenteric mass underwent immunohistochemical staining for IgG (Panel A) and IgG4 (Panel B), with the latter showing diffuse, scattered IgG4-positive plasma cells. The average number of plasma cells that were positive for IgG4 (a mean of 421 cells per high-power field out of three analyses, with a ratio of IgG4-positive plasma cells to IgG-positive plasma cells of 0.53) fit the diagnosis of IgG4-related disease. The renal cortex shows expanded mesangial areas with a loss of argyrophilia and thickened capillary walls (Panel C, silver staining), and apple-green birefringence is visible in the mesangium and along capillaries under polarized light (Panel D, Congo red staining).

the levels of IgG4 (2500 mg per deciliter) and C-reactive protein (8.4 mg per deciliter) and the erythrocyte sedimentation rate (119 mm per hour).

After therapy with prednisone and azathioprine, both the size of the mass and the levels of inflammatory markers decreased rapidly. However, as the prednisone was being tapered, renal impairment with the nephrotic syndrome developed, as indicated by a creatinine level of 1.9 mg per deciliter (168 μ mol per liter), an albumin level of 2.2 g per deciliter, and a ratio of urinary albumin (measured in milligrams per liter) to urinary creatinine (measured in grams per liter) of 10.3. Renal AA amyloidosis was diagnosed on kidney biopsy (Fig. 1C and 1D). Retrospectively, the serum amyloid level from July 2015 had been elevated (350 mg per liter). Treatment with intravenous methylprednisolone and rituximab led to slight decreases in the plasma creatinine level (1.7 mg per deciliter [150 μ mol per liter]) and in the ratio of urinary protein to creatinine (8.2) and a substantial decrease in the levels of both C-reactive protein (0.1 mg per deciliter) and serum amyloid (4 mg per liter).

IgG4-related disease is associated with previously unexplained clinical conditions that include idiopathic retroperitoneal fibrosis.² It is considered to be a fibroinflammatory disease that is responsive to antiinflammatory therapy, which may prevent fibrosis.³ Various long-standing inflammatory conditions such as autoinflammatory and rheumatic disorders may lead to the deposition of AA amyloid.³⁻⁵ Serum amyloid and C-reactive protein are both acute-phase proteins that are often elevated during inflammation. The formation of serum amyloid can be expected in IgG4-related disease, with elevations in C-reactive protein in approximately 40% of patients,² although the incidence

and magnitude of elevated serum amyloid levels remain unknown.

In our patient, the disease appeared to have been dormant for at least 16 years, according to the deduced time that the abdominal mass had been present. However, we speculate that the patient's long-standing inflammatory state probably contributed to the development of amyloidosis. Chronically elevated serum amyloid levels in inflammatory diseases are associated with AA amyloidosis,⁴ but how these levels in patients with IgG4-related disease correlate with organ amyloid depositions is unknown.

In conclusion, AA amyloidosis developed in a patient with IgG4-related disease. It is not known whether the treatment of IgG4-related disease not only ameliorates symptoms but also may modulate inflammatory effects that could potentially lead to secondary amyloidosis.

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Disclosure forms provided by the authors are available with the full text of this letter at NEJM.org.

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***Mycobacterium chimaera* Spread from Heating and Cooling Units in Heart Surgery**

TO THE EDITOR: Recent articles have described patients with infections due to a form of nontuberculous mycobacteria, *Mycobacterium chimaera*, that developed after cardiac surgery.¹⁻³ Infections have been publicly reported in five countries, and it has been noted that these infections developed after the Stöckert 3T heating and cooling unit manufactured by Sorin Group Deutsch-

land (now LivaNova) had been used to regulate the body temperature of patients during coronary-artery bypass grafting.¹⁻⁵ Infections are presumed to be acquired through airborne transmission of aerosolized bacteria from water tanks.¹

After these reports in the Northern Hemisphere of infections associated with heating and cooling units, health care institutions in Australia and